Gonococcal Conjunctivitis — Comprehensive Disease Characterization Report

Disease: Gonococcal Conjunctivitis · **MONDO:** MONDO:0015455 · **Category:** Infectious Disease
Evidence base: Human clinical case series/surveillance, in vitro (human cell) and humanized-mouse mechanistic studies, and recent reviews/RCTs (2005–2026). PMIDs cited throughout.

Summary (Answer to the Research Question)

Gonococcal conjunctivitis is an acute, hyperacute purulent conjunctivitis caused by the obligate human Gram-negative diplococcus ***Neisseria gonorrhoeae***. It occurs in two epidemiologically distinct forms: **adult conjunctivitis/keratoconjunctivitis**, acquired by genital-to-eye autoinoculation or sexual contact (usually unilateral, predominantly young men), and **gonococcal ophthalmia neonatorum (GON)**, acquired by vertical transmission through an infected birth canal. It is a **sight-threatening ophthalmic emergency**: because *N. gonorrhoeae* can invade intact corneal epithelium, delayed treatment leads to rapid stromal melting, corneal perforation, and blindness. The disease is purely infectious (no host genetic causation); pathogenesis proceeds from pilus/LOS/Opa-mediated adhesion to conjunctival columnar epithelium, through CEACAM-receptor-mediated invasion, to intense neutrophil-driven inflammation and protease-mediated keratolysis. First-line treatment is **systemic ceftriaxone** (± azithromycin) plus saline lavage; escalating multidrug resistance (now including ceftriaxone via *penA* alleles) threatens therapy. Prevention rests on safe-sex behavior, STI/prenatal screening, and neonatal ocular prophylaxis; no licensed gonococcal vaccine exists.

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1. Disease Information

Overview. An acute bacterial conjunctivitis caused by *N. gonorrhoeae* characterized by copious purulent discharge, marked conjunctival hyperemia/chemosis, eyelid edema, and a propensity for corneal involvement. Classified as hyperacute bacterial conjunctivitis (onset within ~12–24 h, rapid progression). "Gonococcal conjunctivitis, a diagnostic and therapeutic emergency, is a potentially blinding sexually transmitted disease" ( ). P 24210280

Key identifiers: - **MONDO:** MONDO:0015455 (gonococcal conjunctivitis) - **ICD-10:** A54.31 (gonococcal conjunctivitis); ophthalmia neonatorum P39.1 / A54.3 group - **ICD-11:** 1A72 (Gonococcal infection); ocular involvement under gonococcal infection of eye - **MeSH:** related terms "Gonorrhea" (D006069), "Conjunctivitis, Bacterial" (D003234), "Ophthalmia Neonatorum" (D009878) - **SNOMED CT:** Gonococcal conjunctivitis (disorder) - **OMIM/Orphanet:** Not applicable — this is an acquired infectious disease, not a Mendelian/rare genetic disorder. - **Causative organism:** *Neisseria gonorrhoeae* — **NCBI Taxonomy txid485**

Synonyms / alternative names: gonococcal ophthalmia; gonococcal keratoconjunctivitis (with corneal involvement); gonorrheal conjunctivitis; gonococcal ophthalmia neonatorum / GON (neonatal form); "ophthalmia neonatorum" (when neonatal, of gonococcal etiology); blennorrhea/blennorrhoea neonatorum (historical).

Data source type: Disease-level aggregated resources plus individual-patient case series/case reports and national STI surveillance (EHR-adjacent). No genetic/EHR-cohort molecular dataset applies.

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2. Etiology

Disease causal factor — infectious (single agent). Causative organism is *N. gonorrhoeae*, "the obligate human pathogen [that] colonizes primarily the mucosal columnar epithelium … [and] can infect … epithelial cells of the conjunctiva" ( ). P 31119630

There is **no genetic/heritable cause**: no causal human gene, inheritance pattern, penetrance, or pathogenic germline/somatic variant. ("Causal genes" belong to the pathogen — see §4.)

Transmission / routes. - **Adults:** "Adult gonococcal conjunctivitis (GC) typically occurs by direct inoculation and is usually unilateral" (P 41222259) — genital secretions transferred to the eye (autoinoculation) or via infected sexual fluids. - **Neonates:** vertical transmission during vaginal delivery from an infected maternal genital tract.

Environmental / behavioral risk factors: unprotected sexual intercourse, multiple/new sexual partners, men who have sex with men (MSM), concurrent urogenital/pharyngeal/rectal gonorrhea, sex work, young age (15–34), and (for neonates) untreated maternal gonorrhea and lack of prenatal screening. Male predominance is marked (male:female ≈ 11:1 in gonorrhea surveillance; P 41385198). Bacterial STIs including gonorrhea disproportionately affect MSM and are linked to HIV acquisition (P 41147878).

Genetic risk/modifier factors (host): none causal. At the cellular level, host **CEACAM** receptor expression modulates susceptibility to gonococcal invasion (P 29760215) — a host-factor modifier of infection efficiency, not a disease-risk allele.

Protective factors: condom use, partner reduction, prompt genital hygiene, prenatal STI screening/treatment, and neonatal ocular prophylaxis (behavioral/public-health, not genetic). No known protective host genetic variant.

Gene–environment interactions: Not applicable in the classical human-genetics sense. The relevant "gene–environment" axis is pathogen genotype (resistance/adhesin phase variation) × host receptor repertoire × behavioral exposure.

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3. Phenotypes

Frequencies below are from a 5-year adult culture-confirmed series (n=15) unless noted (P 34211138).

Phenotype	Type	Frequency	HPO suggestion
Purulent (mucopurulent) ocular discharge	Symptom/ sign	93% (14/15)	HP:0031936 (abnormal conjunctiva); (purulent discharge)
Conjunctival hyperemia / injection	Clinical sign	Near-universal	HP:0000509 (Conjunctivitis)
Haemorrhagic conjunctivitis	Clinical sign	67% (10/15)	HP:0000509
Pre-septal / periorbital cellulitis, eyelid edema	Clinical sign	60% (9/15)	HP:0000509 / eyelid edema
Corneal involvement (keratitis)	Clinical sign	33% (5/15)	HP:0000481 (Corneal opacity)/keratitis
Marginal / peripheral ulcerative keratitis	Clinical sign	7% (1/15); P 42305694	HP:0000481
Corneal perforation	Complication	Severe subset; ~11 d post-onset (P 18064057)	HP:0500064 (Corneal perforation)
Chemosis, foreign-body sensation, pain, photophobia, lid swelling	Symptoms	Common	HP:0000509
Pre-auricular lymphadenopathy	Sign	Common	HP:0100807 (Lymphadenopathy)

Characteristics: Onset — hyperacute (hours). **Age** — bimodal: neonatal (days 2–5 of life for GON) and young adults. **Severity** — moderate-to-severe/variable; can rapidly become sight-threatening. **Progression** — rapidly progressive if untreated (branch to keratitis → stromal

melt → perforation); treatment-induced remission is typical and complete. **Laterality** — usually **unilateral** in adults (autoinoculation), **bilateral** in neonates (and occasionally severe adult cases,).

P 41222259

Quality-of-life impact: acute painful red eye with functional impairment; sight-threatening in the keratitis subset. At population level, gonorrhea causes substantial QALY loss (per-infection QALY estimates highest in women;),

P 36806950

though ocular-specific QoL instruments are not established.

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4. Genetic / Molecular Information

Not a human genetic disease — no causal human gene, pathogenic variant (ACMG/AMP), allele frequency, or chromosomal abnormality. The "molecular" determinants are **pathogen** genes and **host receptors**:

Pathogen virulence genes (*N. gonorrhoeae*): - **pilE / pilin (type IV pili)** — initial attachment; antigenic/phase variation. - **opa (opacity-associated Opa proteins)** — phase-variable adhesins binding host CEACAMs (

P 27442026

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P 21204865

- **porB (porin PorB)**, **iron-regulated proteins (e.g., TbpB)**, **lgtA–E (LOS biosynthesis)**, secreted enzymes including **IgA1 protease** ( ).

P 16532656

- **Resistance determinants:** ***penA*** (mosaic allele 60.001 → ceftriaxone resistance;),

P 42430114

penicillinase plasmid (blaTEM, PPNG) ( ),

P 42475726

tet(M), *gyrA/parC* (fluoroquinolone resistance), *mtrR* efflux, *23S rRNA/mosaic mtr*

(azithromycin). - "Instability of gonococcal antigens, and frequent formation of new antigenic variants ... [is] the main cause of the absence of postinfectious immunity" (P 16532656).

Host receptor/molecular players: CEACAM1, CEACAM3, CEACAM5 (CEA), CEACAM6 (HGNC gene symbols CEACAM1/3/5/6); differential CEACAM expression determines infection outcome (P 29760215). P 17517873

Epigenetics / chromosomal abnormalities: Not applicable to host. (Bacterial phase variation is mediated by slipped-strand mispairing at homopolymeric/repeat tracts — a pathogen genetic mechanism.)

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5. Environmental Information

- **Infectious agent:** *Neisseria gonorrhoeae* (NCBI:txid485), Gram-negative diplococcus, family Neisseriaceae. Frequent co-infection with *Chlamydia trachomatis* (P 41017068).
- **Environmental/occupational toxins, radiation, pollution:** not implicated.
- **Lifestyle factors:** high-risk sexual behavior is the dominant modifiable factor (unprotected sex, multiple partners, MSM networks, transactional sex); alcohol/drug use (e.g., chemsex) as behavioral amplifiers. Neonatal disease reflects maternal infection and absent prenatal care/screening.

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating infection → clinical manifestation)

1. **Inoculation** of *N. gonorrhoeae* onto the **conjunctival columnar epithelium** (autoinoculation, sexual contact, or birth-canal passage) → **leads to** surface colonization

(demonstrated)
(P 31119630).

2. **Type IV pili** mediate initial long-range attachment and twitching motility; **LOS** and **Opa/porin** promote intimate adherence → **results in** stable epithelial binding (demonstrated in vitro).
3. **Phase-variable Opa proteins engage host CEACAM receptors** (CEACAM1/5/6 on epithelium) → **triggers** bacterial uptake/invasion: epithelial CEACAMs drive **cholesterol-raft-dependent** internalization, whereas granulocyte **CEACAM3** drives **Src-family-kinase-dependent** phagocytosis (demonstrated)
(P 17517873). P 27442026
Branch: "Opa binding to CEACAM1 on … epithelia enhances gonococcal association and penetration into these tissues"
(P 29760215).
4. Epithelial invasion and transcytosis → **leads to** subepithelial bacterial access and release of LOS/peptidoglycan (PAMPs) → **activates** innate immune signaling (TLR4/lipid A, NOD1/peptidoglycan; *inferred* for conjunctiva by analogy to genital mucosa) → **results in** cytokine/chemokine release.
5. Chemokine gradients → **drive** massive **neutrophil (PMN) recruitment**: "N. gonorrhoeae infection is characterized by local and abundant recruitment of neutrophils"
(P 35343795)
→ **produces** the hallmark **purulent exudate**.
6. *N. gonorrhoeae* **survives within neutrophils** via phase-variable Opa expression and antioxidant defenses → **sustains** persistent inflammation (demonstrated)
(P 35343795). P 25346239
7. Neutrophil and bacterial **proteases / collagenolytic activity** plus oxidative injury → **cause** corneal **stromal keratolysis ("stromal melting")** (mechanistically inferred from clinical

rapidity)
(P 32787586).

8. Progressive stromal destruction → **leads to corneal perforation** (mean ~11 days post-onset if untreated;
(P 18064057)
→ **results in** loss of vision/globe integrity (clinical manifestation).
9. In parallel, **antigenic/phase variation** of pili and Opa → **enables** immune evasion → **results in** absence of protective immunity and repeated susceptibility
(P 16532656).

Mechanistic detail (checklist mapping)

- **Molecular pathways:** Src-family-kinase (SFK) signaling downstream of CEACAM3
(P 17517873);
cholesterol-rich membrane microdomain/lipid-raft uptake for epithelial CEACAMs; innate PRR signaling (TLR4-MD2, NOD1) inferred.
- **Cellular processes:** bacterial adhesion (GO:0007155), receptor-mediated endocytosis/phagocytosis (GO:0006909), inflammatory response (GO:0006954), leukocyte chemotaxis/migration (GO:0050900), respiratory burst (GO:0045730).
- **Protein dysfunction:** host — none intrinsic; pathogen exploits normal CEACAM adhesion function. Antigenic variation of Opa/pilin.
- **Immune involvement:** acute neutrophilic inflammation; incomplete/no adaptive immunity; complement evasion
(P 21204865).
- **Tissue damage mechanisms:** protease-mediated collagen degradation, oxidative stress, necrosis of corneal stroma → keratolysis/perforation.
- **Molecular profiling (omics):** No disease-specific human transcriptomic/proteomic/metabolomic signatures of gonococcal conjunctivitis were identified; mechanistic data derive from in vitro cell models and humanized mice.

Cell types (CL): neutrophil (CL:0000775), conjunctival/corneal epithelial cell (CL:0000066), granulocyte. **GO components:** plasma membrane raft (GO:0045121), phagolysosome (GO:0032010).

7. Anatomical Structures Affected

- **Primary organ/site:** the **eye** — specifically the **conjunctiva** (UBERON:0001811), palpebral and bulbar; eyelids/lid margin (UBERON:0000441).
- **Secondary involvement: cornea** (UBERON:0000964) — epithelium and stroma (keratitis, ulceration, stromal melt, perforation); periorbital soft tissue (pre-septal cellulitis); pre-auricular lymph nodes. Endophthalmitis/panophthalmitis in extreme untreated cases.
- **Body systems:** ocular/visual system (primary); reproductive/genitourinary system (source reservoir); rarely musculoskeletal/systemic via **disseminated gonococcal infection (DGI)** — arthritis-dermatitis syndrome, septic arthritis
  (P 42273458).
- **Tissue types:** stratified/columnar **epithelial tissue** (conjunctiva, cornea); corneal **connective tissue (stroma)**.
- **Cell populations (CL):** conjunctival epithelial cells, corneal epithelial cells (CL:0000575), corneal stromal keratocytes (CL:0002363), infiltrating neutrophils (CL:0000775).
- **Subcellular (GO CC):** plasma membrane / lipid raft (GO:0045121); phagosome–phagolysosome (GO:0045335/GO:0032010).
- **Localization / laterality:** adults usually **unilateral**; neonates typically **bilateral**; severe adult disease may be bilateral
  (P 41222259).

8. Temporal Development

- **Onset: hyperacute** (symptoms within ~12–24 h of inoculation). Age: **neonatal** GON typically presents days **2–5** of life; **adult** form in sexually active young adults.
- **Progression:** rapid. Untreated adult keratoconjunctivitis can progress to **corneal perforation at a mean of ~11 days** after conjunctivitis onset
  (P 18064057).

Frequent early misdiagnosis as viral/epidemic keratoconjunctivitis delays therapy (P 18064057).

- **Disease course:** acute, **self-limited to the episode** with treatment; not chronic or relapsing (though reinfection can occur). Treatment-induced remission is the norm — "a complete cure was achieved without sequelae" in an uncomplicated case (P 24210280).
- **Critical period:** the **days-to-~1-week window** before perforation is the key opportunity for sight-saving intervention; prompt systemic antibiotics + lavage prevent progression.

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9. Inheritance and Population (Epidemiology)

- **Inheritance: None** — acquired infection; no Mendelian pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity, or carrier frequency applies.
- **Global burden of gonorrhea: >80 million new cases annually** worldwide (P 42773319).

  Ocular gonococcal infection is an **uncommon extragenital manifestation** but is **rising** with the STI resurgence: a 20.5-year Zurich series found "a marked increase in gonococcal cases … between 2021 and 2025" (17 gonococcal vs 121 chlamydial ocular cases) (P 42795570).
- **Incidence example (gonorrhea overall):** Poland 2023 — incidence doubled to **3.5/100,000** (1,322 cases), still far below EU/EEA average (~25/100,000); adult conjunctival infections rose to **1%** of cases (vs 0.3% in 2022) (P 41385198).

  US county-level gonorrhea rates declined 2021–2023 in ~74% of counties (P 42641714).
- **Sex ratio:** strongly **male-predominant** for gonorrhea (11:1 in Poland; 80% male in the Glasgow ocular series)

(P 41385198). P 34211138

- **Age distribution:** concentrated in **15–34-year-olds** (mean 26 in the ocular series); plus **neonates**.
- **Geographic:** highest burden in sub-Saharan Africa, Western Pacific, and among key populations globally; ceftriaxone-resistant lineages expanding in Asia (e.g., ST8123 in Hangzhou, China;

). P 42430114

- **Neonatal GON:** now **rare in high-income countries** — an Italian survey of 419 centers (>1 million neonates/3 y) reported **zero** gonococcal (and 12 chlamydial) neonatal conjunctivitis cases

(P 37697419).

10. Diagnostics

- **Gram stain** of conjunctival purulent exudate: **Gram-negative intracellular diplococci** within neutrophils (rapid presumptive diagnosis)

(P 24210280). P 32787586

- **Culture:** modified Thayer-Martin / chocolate agar; enables **antimicrobial susceptibility testing** (essential given resistance)

(P 34211138).

- **NAAT / PCR:** e.g., **Xpert CT/NG** provides same-day diagnosis and can be run on genital or ocular swabs; "Speedy and accurate diagnosis by the highly sensitive and specific Xpert CT/NG assay can provide early definite diagnosis for prompt treatment in prevention of gonococcal infection induced corneal perforations"

(P 32787586).

Caveat: "No regulatory agency-approved nucleic acid amplification tests (NAATs) exist"

specifically for ocular gonococcal/chlamydial detection — used off-label
(P 42429409).

- **Whole-genome sequencing:** increasingly used for resistance surveillance/typing (penA alleles, MLST)
(P 42430114).
- **Clinical criteria / differential diagnosis:** hyperacute copious purulent unilateral conjunctivitis in a sexually active adult ⇒ suspect gonococcus. Differentials: **chlamydial (inclusion) conjunctivitis** (more subacute/chronic; the main mimic, P 42795570), adenoviral epidemic keratoconjunctivitis (watery discharge — a frequent misdiagnosis; P 18064057), other bacterial (staph/strep/Haemophilus), and (neonates) chemical conjunctivitis.
- **Ancillary:** because ocular gonorrhea signals systemic STI, **concurrent testing for chlamydia, syphilis, HIV** and **partner tracing** is standard
(P 34211138).
- **Screening:** prenatal maternal gonorrhea/chlamydia screening (secondary prevention of GON); high-risk-population STI screening. No newborn/genetic screening applies.
- **Omics/imaging/electrophysiology/biopsy:** generally not used for diagnosis (clinical + microbiology suffice).

LOINC-type tests: *N. gonorrhoeae* culture; *N. gonorrhoeae* rRNA/DNA NAAT; Gram stain.

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11. Outcome / Prognosis

- **With prompt systemic treatment:** excellent — complete cure without sequelae in uncomplicated cases
(P 24210280).

- **With delay/keratitis:** sight-threatening. Corneal perforation develops ~11 days post-onset in severe cases (P 18064057).
- **Surgical salvage:** therapeutic keratoplasty (lamellar/penetrating/tectonic) plus systemic antibiotics restores vision — in a 5-case series "All grafts remained clear during the mean follow-up period of 34.9 months. Final best-corrected visual acuity ranged from 20/60 to 20/16" (P 18064057); bilateral perforation salvaged by tectonic penetrating keratoplasty (P 41222259).
- **Mortality:** ocular disease itself is essentially non-fatal; morbidity is **visual** (corneal scarring, blindness) if untreated. Systemic DGI carries morbidity (septic arthritis; P 42273458).
- **Prognostic factors:** time-to-treatment (dominant), presence/depth of corneal involvement, antimicrobial susceptibility (resistant strains worsen outcomes — 3/5 perforation cases were ofloxacin-resistant; P 18064057), immune status, and diagnostic delay/misdiagnosis.
- **Complications:** keratitis, corneal ulcer/perforation, peripheral ulcerative keratitis (P 42305694), pre-septal cellulitis, endophthalmitis, and (rarely) DGI.

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12. Treatment

Pharmacotherapy (systemic is mandatory — topical alone insufficient because of corneal invasion risk): - **First-line: ceftriaxone** (third-generation cephalosporin; inhibits penicillin-binding proteins/cell-wall synthesis), typically IM/IV; adult ocular disease treated with **IV or IM ceftriaxone**, often with **azithromycin** (P 34211138).

CHEBI: ceftriaxone (CHEBI:29007); NCIT: Ceftriaxone (C1145). - **Adjunct: azithromycin** 1 g (macrolide; historically for dual therapy / chlamydia coverage). CHEBI:2955; NCIT: Azithromycin (C574). - **Ocular supportive care: copious saline lavage** of purulent discharge; topical antibiotics as adjunct; topical lubrication (P 34211138). P 24210280

- **Novel oral agents (FDA-approved Dec 2025): zoliflodacin** (spiropyrimidinetrione, targets bacterial type II topoisomerase) and **gepotidacin** (triazaacenaphthylene topoisomerase inhibitor) for uncomplicated urogenital gonorrhea — "obtained US FDA-approval for treatment of uncomplicated urogenital gonorrhea in December 2025" (P 42728217).

Antimicrobial resistance (defines therapy): *N. gonorrhoeae* "has developed resistance to all antibiotics recommended for treatment, and reports of reduced susceptibility to ceftriaxone, the last-line treatment, are increasing" (P 39673766).

Sequential loss of sulfonamides → penicillins (PPNG) → tetracyclines → fluoroquinolones → emerging cephalosporin resistance (**penA allele 60.001**; FC428/ST1903 and endemic ST8123 lineages; high-level ceftriaxone MIC ≥0.5 mg/L reaching 35% in 2024 Hangzhou) (P 42430114). P 42475726

Surgical/interventional: therapeutic/tectonic **keratoplasty** for perforation (NCIT: Corneal Transplantation C15326) (P 18064057). P 41222259

Pharmacogenomics / advanced therapeutics (gene/cell/RNA/targeted/immunotherapy): not applicable to treatment of the infection.

Treatment strategy: empiric systemic ceftriaxone on clinical/Gram-stain suspicion, with culture-guided adjustment; test-of-cure and partner treatment; management shifting toward "etiology-guided and AMR-informed therapy, driven by advances in rapid point-of-care testing and whole-genome sequencing" (P 42728217).

13. Prevention

- **Primary prevention:** safe-sex behavior (condoms), partner reduction, and sexual-health education; **neonatal ocular prophylaxis** at birth — historically **Credé silver nitrate**, now **erythromycin 0.5% ophthalmic ointment** (mandated in many US states) or **povidone-iodine** (WHO-favored in some settings). NCIT: erythromycin ophthalmic ointment; silver nitrate; povidone-iodine.
- **Secondary prevention: prenatal maternal STI screening and treatment** — "Prenatal screening and treatment of pregnant women is the most effective way to prevent neonatal ophthalmia" (P 38781483); high-risk-population STI screening/treatment; prompt treatment to prevent transmission.
- **Tertiary prevention:** rapid diagnosis + systemic antibiotics + lavage to prevent keratitis/perforation.
- **Policy evolution:** universal neonatal erythromycin prophylaxis is being reconsidered — GON is "exceedingly uncommon" in the US and erythromycin is ineffective for chlamydial ophthalmia; risk-based models are favored where prenatal screening is robust (P 41344385). P 38781483 Breakthrough neonatal GON still occurs with resistant strains despite prophylaxis (P 41017068).
- **Vaccine: no licensed gonococcal vaccine** (P 42773319).

The meningococcal **4CMenB** vaccine (shares antigens with gonococcus) shows **~30–38% cross-protection in observational studies** (meta-analysis 38% risk reduction) and is deployed for high-risk groups in the UK and Galicia, Spain — but **three recent RCTs (incl. GoGoVax) showed no efficacy** against gonorrhoea in high-risk MSM
(P 42499182). P 42773319
NCIT: 4CMenB vaccine.

- **Public health:** partner notification/contact tracing, STI surveillance, and antimicrobial stewardship/resistance monitoring.

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14. Other Species / Natural Disease

- **Taxonomy of pathogen:** *Neisseria gonorrhoeae* — **NCBI:txid485**. Host species: ***Homo sapiens* (NCBI:txid9606) only.**
- **Natural animal disease: None** — *N. gonorrhoeae* is an **obligate/strict human pathogen** with **no natural animal reservoir**
(P 31119630).
No veterinary counterpart; **not zoonotic**. VBO/OMIA: not applicable.
- **Comparative biology:** the closest relative is *Neisseria meningitidis*, which shares adhesins and Opa-CEACAM biology; cross-protective antigens underlie 4CMenB cross-immunity
(P 42499182).
Host **CEACAM** genes have orthologs across mammals, but human-specific CEACAM sequences restrict gonococcal Opa binding to humans (basis for humanized-mouse models).
- **Cross-species susceptibility:** effectively none under natural conditions.

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15. Model Organisms

- **No natural animal model** (human-restricted pathogen), so research relies on **surrogate systems** (P 31119630).
- **Humanized mouse: CEACAM-humanized transgenic mice** (expressing human CEACAM1/CEACAM5) recapitulate Opa-CEACAM-dependent epithelial colonization and tissue penetration (P 29760215). The **estradiol-treated female mouse genital-tract colonization model** is the standard in vivo infection model (used widely for vaccine/therapeutic testing; general field knowledge).
- **In vitro / cellular:** human epithelial cell monocultures and **primary human neutrophils** to study adhesion, invasion, and phagocytic killing (P 31119630 P 35343795); P 25346239 "protocols for examining the basic interactions between gonococci and a representative human cell line" (P 31119630).
- **Genetic model types:** transgenic/humanized (CEACAM knock-in) mice; no knockout-of-host-gene disease models (disease is infectious).
- **Applications:** dissecting adhesin–receptor biology, neutrophil evasion, and testing vaccine antigens — e.g., murine immunization/challenge with **TbpB** (P 42574476), **BamA mRNA-LNP** (P 42523399), and **native outer membrane vesicle** vaccines (P 42363875); a controlled **human urethral challenge model** (male volunteers) also exists for vaccine

efficacy testing (e.g., 4CMenB trial NCT05294588;
).
P 42049314

- **Limitations:** models do not fully reproduce the **human conjunctival/corneal** microenvironment; mouse immune and CEACAM differences limit translation; no dedicated ocular animal model of gonococcal keratoconjunctivitis.
- **Resources:** IMSR/MGI (transgenic mouse strains), Cellosaurus/ATCC (human epithelial cell lines).

Supported vs. Refuted Hypotheses

Supported: - H1 (supported): Adult gonococcal conjunctivitis is a hyperacute, male-predominant, usually unilateral purulent conjunctivitis with frequent corneal/pre-septal involvement
(
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P 34211138
- H2 (supported): Corneal stromal melt/perforation is the principal sight-threatening complication, driven by epithelial invasion + neutrophilic protease injury
(
P 32787586
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P 18064057
- H3 (supported): Opa-CEACAM interactions mechanistically drive epithelial invasion and modulate neutrophil handling
(
P 29760215
P 17517873
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P 35343795
- H4 (supported): Systemic ceftriaxone is first-line; multidrug (incl. ceftriaxone) resistance is escalating
(
P 39673766
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P 42430114

- H5 (supported): Prevention hinges on prenatal/behavioral measures; GON is now rare in high-income settings (P 38781483). P 37697419

Refuted / not applicable: - Gonococcal conjunctivitis has a heritable/host-genetic cause — **refuted**: purely infectious; no causal human gene, inheritance, or pathogenic variant. - 4CMenB reliably prevents gonorrhea — **refuted by RCTs** (effective only ~30–38% in observational data, no efficacy in MSM RCTs) (P 42499182). - A natural animal reservoir/model exists — **refuted**: human-restricted pathogen (P 31119630).

Limitations and Future Directions

- **Evidence base** for the ocular form is dominated by **case series/reports** (small n) and surveillance; no large ocular-specific RCTs or omics datasets exist. Mechanistic data are extrapolated from genital-tract/in-vitro models.
- **Ontology gaps:** some HPO/NCIT mappings are approximate; exact conjunctival-purulent-discharge HPO code is imprecise.
- **Key uncertainties:** trajectory of ceftriaxone resistance and its impact on ocular disease; whether next-gen vaccines (OMV, mRNA) will protect mucosal/ocular sites; optimal neonatal prophylaxis policy in the screening era.
- **Future work:** conjunctival organoid/humanized-cornea models; ocular-specific resistance surveillance; evaluation of gepotidacin/zoliflodacin and vaccine candidates for ocular/mucosal protection.

Evidence source legend: human clinical (34211138, 24210280, 18064057, 41222259, 42305694, 34211138, 42795570, 41385198, 42641714, 37697419, 41017068, 42273458); in vitro/human-cell (31119630, 35343795, 25346239, 17517873, 27442026); humanized mouse (29760215); review/

RCT (42728217, 39673766, 42430114, 42475726, 42773319, 42499182, 38781483, 41344385, 16532656, 21204865, 42574476, 42523399, 42363875, 42049314); population/health-economics (36806950, 41147878).
